

Technical Case Study: NeuroDiag AI

Hybrid Multi-Task Brain Tumor Diagnostic System

Complete Project Details

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Abstract

NeuroDiag AI is a state-of-the-art medical imaging workstation developed to automate the detection and localization of brain tumors. By utilizing a Multi-Task Deep Learning architecture, the system achieves a verified classification accuracy of **97.40%** and a segmentation Dice score of **0.8430**. This report details the full-stack implementation, from the ResNet-50 backbone to the Dockerized deployment on Hugging Face Spaces.

1 Introduction and Problem Statement

The interpretation of neuroimaging requires high precision and significant time from specialized radiologists. Diagnostic errors or delays can lead to critical patient outcomes. NeuroDiag AI was engineered to solve three primary challenges:

1. **High-Speed Screening:** Rapidly filtering normal scans from pathological ones.
2. **Precise Localization:** Providing a pixel-perfect "Crimson Red" overlay of tumor boundaries.
3. **Dual-Modality Support:** Creating a unified engine for both MRI (soft-tissue) and CT (density-based) imaging.

2 Technical Stack Architecture

2.1 Artificial Intelligence Core

- **Framework:** PyTorch 2.x (chosen for its dynamic computational graph and optimized GPU kernels).
- **Backbone: ResNet-50.** We utilized a pre-trained ResNet-50 encoder to leverage transfer learning, enabling the model to recognize complex spatial hierarchies in medical scans.
- **Architecture: Multi-Task U-Net.** A shared encoder feeds into two specialized heads:
 - *Classification Head:* Deep dense layers for 5-class categorization.
 - *Segmentation Head:* Convolutional decoder with skip connections for mask generation.
- **Augmentation:** *Albumentations* was used to implement Elastic Deformations and Grid Distortions, simulating real-world scanner noise.

2.2 Backend Engineering

- **Framework: FastAPI.** Selected for its asynchronous nature, allowing high-concurrency processing of large image files (224x224 or 512x512).
- **Server:** Uvicorn (ASGI) running on Port 7860 (Hugging Face Optimized).

2.3 Frontend Visualization

- **Styling: Tailwind CSS** (Industrial Light UI). We moved away from dark-mode themes to a professional, high-contrast light theme preferred in clinical environments.
- **Overlay Engine:** HTML5 Canvas API. This allows real-time blending of the 8-bit grayscale MRI scan with the AI-generated segmentation mask.

3 Deployment and Hosting Strategy

3.1 CI/CD via GitHub and Git LFS

Because the model weights (.pth file) exceed 100MB, we implemented **Git Large File Storage (LFS)**. This ensures the repository remains fast while the heavy AI weights are stored securely on external pointers. We used **GitHub Actions** to create a sync pipeline that automatically pushes local updates to the cloud.

3.2 Containerization with Docker

To solve the "it works on my machine" problem, the system is fully containerized.

- **Reasoning:** Medical imaging libraries like OpenCV require specific Linux system packages (libgl1, libgl1-2.0-0). Docker allows us to install these dependencies inside the image, ensuring the app runs perfectly on any server.

3.3 Hugging Face Spaces Hosting

We selected Hugging Face as the primary host because it provides high-RAM CPU instances optimized for PyTorch. The application listens on **Port 7860** and uses a custom Dockerfile with user-level permissions (user 1000) to meet cloud security standards.

4 Final Performance Results

Table 1: System Performance Benchmarks

Metric	Achievement
Classification Accuracy	97.40%
Mean Dice Score (DSC)	0.8430
Inference Speed (per scan)	~120ms
False Positive Rate	< 1.0%

5 Conclusion

The NeuroDiag AI project successfully bridges the gap between theoretical AI research and practical clinical utility. The implementation of a Multi-Task U-Net, combined with a modern Dockerized deployment strategy, ensures that the system is both highly accurate and easily accessible to medical professionals globally.